06 Comparators - Rule-Based, Cox, and GBT Comparator Models

Abdul Ahad. Student ID: G00486649

31 August 2026

Table of Contents

0.1 Introduction 1
0.2 Feature Set 1
0.3 Fit Comparators Per Variant 1
0.4 Ablated GBT (Protocol Amendment 5) 2
0.5 Random-Split Comparators (H4) 3
0.6 Feature Importance (GBT) 3

0.1 Introduction

All comparators share the same cohort, labels, features, temporal split, and metrics as the primary model (Reyna et al., 2020).

Comparator	Rationale
NEWS2	Rule-based sepsis screening score (Evans et al., 2021). Pre-computed.
qSOFA, SIRS	Descriptive context only.
Static Cox PH	Weak baseline with time-fixed covariates. PH assumption expected to be violated.
XGBoost	ML comparator per PhysioNet conventions (Reyna et al., 2020). Tests H6.

Variant B is the primary comparison; A and C feed the variance decomposition.

0.2 Feature Set

Same feature set as the primary model, and the identity is load-bearing. H6 asks whether the model class matters once everything else is fixed, so the comparators have to see exactly the columns the primary model saw. Give XGBoost a richer input and any gap it opens up is a difference in features wearing the costume of a difference in learners.

0.3 Fit Comparators Per Variant

Per variant: NEWS2/qSOFA/SIRS scores are read from the person-hour table (no fitting); Cox is fitted with Schoenfeld testing; XGBoost uses class-imbalance weighting, an AUCPR evaluation metric, and early stopping on a 15% stay-level holdout carved out of the *training* rows (Protocol Amendment 8). The test set is never in the watchlist.

```
##   Feature check: all 22 declared features present.
##     NEWS2 alert rate: 47.2%
##   Cox PH test: global p = 0.000 (p < 0.05 = PH violated, expected)
##   GBT early-stopping split [full/A]: 45004 fitting stays / 7942 held-out stays (15.0%); event
##   Comparator predictions saved for Variant A
##   Feature check: all 22 declared features present.
##     NEWS2 alert rate: 47.3%
##   Cox PH test: global p = 0.000 (p < 0.05 = PH violated, expected)
##   GBT early-stopping split [full/B]: 45004 fitting stays / 7942 held-out stays (15.0%); event
##   Comparator predictions saved for Variant B
##   Feature check: all 22 declared features present.
##     NEWS2 alert rate: 45.5%
##   Cox PH test: global p = 0.000 (p < 0.05 = PH violated, expected)
##   GBT early-stopping split [full/C]: 45004 fitting stays / 7942 held-out stays (15.0%); event
##   Comparator predictions saved for Variant C
##   Feature check: all 22 declared features present.
##     NEWS2 alert rate: 41.6%
##   Cox PH test: global p = 0.000 (p < 0.05 = PH violated, expected)
##   GBT early-stopping split [full/D]: 45004 fitting stays / 7942 held-out stays (15.0%); event
##   Comparator predictions saved for Variant D
##   Feature check: all 22 declared features present.
##     NEWS2 alert rate: 43.4%
##   Cox PH test: global p = 0.000 (p < 0.05 = PH violated, expected)
##   GBT early-stopping split [full/E]: 45004 fitting stays / 7942 held-out stays (15.0%); event
##   Comparator predictions saved for Variant E
##   Feature check: all 22 declared features present.
##     NEWS2 alert rate: 47.0%
##   Cox PH test: global p = 0.000 (p < 0.05 = PH violated, expected)
##   GBT early-stopping split [full/F]: 45004 fitting stays / 7942 held-out stays (15.0%); event
##   Comparator predictions saved for Variant F
```

0.4 Ablated GBT (Protocol Amendment 5)

The gradient-boosted comparator is re-fitted without the action-derived features, so the ablation is measured on both a linear and a flexible learner. Only XGBoost is ablated: NEWS2, qSOFA and SIRS never used these features, and the Cox comparator is a static single-row-per-stay specification whose role in the thesis is the proportional-hazards check rather than discrimination.

```
##   Feature check: all 22 declared features present.
##   Ablated arm: 14 features (from 22)
##   GBT early-stopping split [ablated/A]: 45004 fitting stays / 7942 held-out stays (15.0%); ev
##   Feature check: all 22 declared features present.
##   Ablated arm: 14 features (from 22)
##   GBT early-stopping split [ablated/B]: 45004 fitting stays / 7942 held-out stays (15.0%); ev
##   Feature check: all 22 declared features present.
##   Ablated arm: 14 features (from 22)
##   GBT early-stopping split [ablated/C]: 45004 fitting stays / 7942 held-out stays (15.0%); ev
```

Table 1: Top 20 GBT features by Gain (Variant B)

Feature	Gain	Cover	Frequency
temp_c	0.1446	0.1442	0.0827
resp_rate_slope6h	0.1375	0.0854	0.0752
map_slope6h	0.0872	0.0980	0.0678
hr_slope6h	0.0831	0.0688	0.0840
gcs	0.0724	0.0687	0.0551
wbc	0.0653	0.0858	0.0775
creatinine	0.0578	0.0761	0.0648
hr	0.0541	0.0612	0.0847
bilirubin_total	0.0500	0.0595	0.0609
map	0.0497	0.0377	0.0840
lactate	0.0438	0.0465	0.0542
platelets	0.0430	0.0468	0.0654
resp_rate	0.0383	0.0350	0.0600
spo2	0.0197	0.0244	0.0378
lactate_measured	0.0147	0.0080	0.0104
norepi_epi_any	0.0126	0.0233	0.0089
vaso_any	0.0085	0.0120	0.0082
pf_ratio_measured	0.0075	0.0096	0.0110
bilirubin_total_measured	0.0056	0.0056	0.0039
creatinine_measured	0.0028	0.0016	0.0015

0.5 Random-Split Comparators (H4)

GBT and rule-based comparators are re-fitted on the random patient-level split for Variant B, so H4 can compare temporal vs. random AUROC across all models.

```
##   Feature check: all 22 declared features present.
##   GBT early-stopping split [random/B]: 41487 fitting stays / 7321 held-out stays (15.0%); eve
##   Random-split comparator predictions saved for Variant B
```

0.6 Feature Importance (GBT)

GBT feature importance provides a cross-check: if GBT relies on the same features as the multinomial regression, both model classes are learning similar structure.